

# Simulation and Modeling

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Lecture 09

## **Random Variate Generations**

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# Convolution Method

- RV  $X$  is a sum of IID random variables

$$X = Y_1 + Y_2 + \cdots + Y_n$$

- A closed form solution for the inverse of  $F_X(x)$  might not exist
- Instead of generating  $X$  directly, by generating  $Y_i$ 's individually, and  $X$  is generated by summing them
  - The term convolution is used from stochastic process
  - $m$ -fold convolution of the distribution of  $Y_i$ 's

# Convolution Method: Algorithm

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- Generate  $u_1, u_2, \dots, u_n$
- Generate  $Y_1, Y_2, \dots, Y_n$  using the inverse transform method
- Calculate  $X = Y_1 + Y_2 + \dots + Y_n$
- Return  $X$

# Example: $m$ -Erlang Distribution

- $m$ -Erlang random variable with mean  $\beta$ 
  - Sum of  $m$  IID exponential random variables with common mean  $\frac{\beta}{m}$
  - $X = \sum_{i=1}^m X_i$
  - Using inverse transform method,  $X_i = -\frac{\beta}{m} \ln(1 - u_i)$
  - Then, 
$$\begin{aligned} X &= \sum_{i=1}^m -\frac{\beta}{m} \ln(1 - u_i) \\ &= -\frac{\beta}{m} \sum_{i=1}^m \ln(1 - u_i) \\ &= -\frac{\beta}{m} \ln[\prod_{i=1}^m (1 - u_i)] \\ &= -\frac{\beta}{m} \ln[(1 - u_1) \times (1 - u_2) \times \cdots \times (1 - u_m)] \end{aligned}$$

# Example: $m$ -Erlang Distribution

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- Algorithm
  - Step 1: Generate  $u_1, u_2, \dots, u_n$
  - Step 2: Calculate  $X = -\frac{\beta}{m} \ln[\prod_{i=1}^m (1 - u_i)]$
  - Step 3: Return  $X$

# Acceptance-Rejection Method

- Generate random variate  $X$  from CDF  $F(x)$  with corresponding PDF  $f(x)$
- However, instead of from  $f(x)$ , we generate  $X$  from another distribution  $g(x)$ , and
- We accept  $X$  with a probability proportional to  $\frac{f(X)}{g(X)}$
- Otherwise, we reject  $X$  and repeat the procedure

# Acceptance-Rejection Method

- Algorithm: Generate random variates,  $X$ , uniformly distributed between  $\frac{1}{4}$  and 1.
  - Step 1: Generate random number  $u_i$
  - Step 2: if  $u_i \geq \frac{1}{4}$ , accept  $u_i$  as  $x_i$ 
    - If another random variate is needed go to Step 1
    - Otherwise, stop.
  - Step 3: If  $u_i < \frac{1}{4}$ , reject  $u_i$ , and go to Step 1.
- Step 2 is an acceptance and Step 3 is a rejection

# Acceptance-Rejection Method

- Accepted values  $u_i$ 's are conditioned values
  - $u_i$ 's do not have the desired distribution
  - But conditioned on the event  $u_i \geq \frac{1}{4}$ , it does have the desired distribution
- To show this, assume  $\frac{1}{4} \leq a < b \leq 1$

$$P \left[ a < u \leq b \middle| \frac{1}{4} \leq u \leq 1 \right] = \frac{P[a < u \leq b]}{P \left[ \frac{1}{4} \leq u \leq 1 \right]} = \frac{b - a}{\frac{3}{4}}$$

- This is a correct probability for uniform distribution on  $[\frac{1}{4}, 1]$

# Acceptance-Rejection Method

- Generate random variate  $X$  with CDF  $F(x)$  and PDF  $f(x)$
- A function  $g(x)$  is specified, called the majoring function, such that
  - $g(x) \geq f(x)$
- $g(x)$  will not, in general, be a PDF, since
  - $c = \int_{-\infty}^{+\infty} g(x)dx \geq \int_{-\infty}^{+\infty} f(x)dx = 1$
- Therefore, we find a function,  $r(x)$ , which is a PDF
  - $h(x) = \frac{g(x)}{c}$
  - Clearly,  $\int_{-\infty}^{+\infty} h(x) dx = 1$
- We generate random variate,  $y$ , with PDF  $h(x)$ 
  - We accept  $y_i$ , when it satisfies certain condition
    - It has the same distribution as  $f(x)$
    - When a random number  $u_i$  satisfies,  $u_i \leq \frac{f(y)}{g(y)}$
  - Otherwise,  $y$  is rejected as a random variate of PDF  $f(x)$

# Acceptance-Rejection Method

- Algorithm
  - Step 1: Generate  $Y$  having PDF  $h(x)$
  - Step 2: Generate  $u \sim U(0, 1)$ , independent of  $Y$
  - Step 3: If  $u \leq \frac{f(Y)}{g(Y)}$ , then  $X = Y$ 
    - Else, go back to Step 1.

# Example 1: Acceptance-Rejection

- Generate random variates for

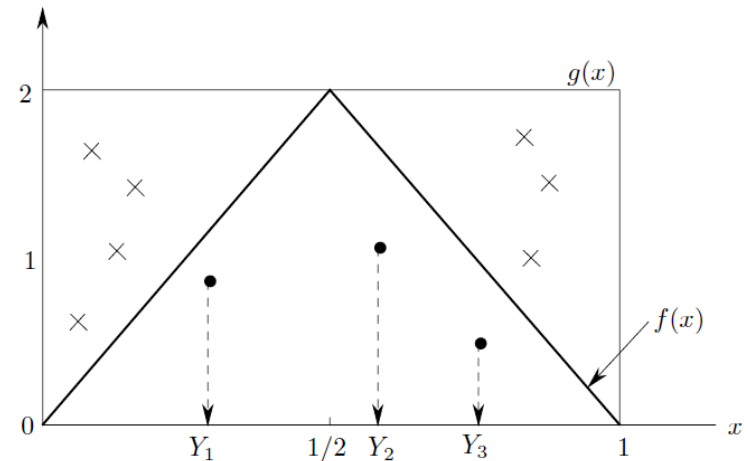
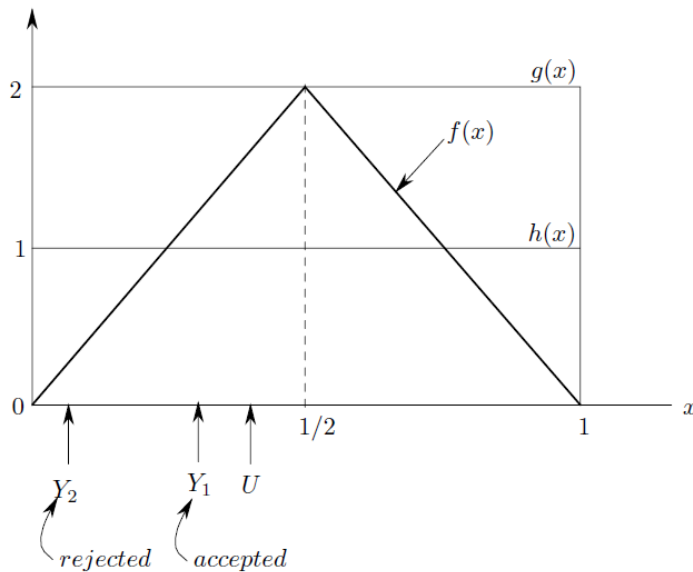
$$f(x) = \begin{cases} 4x, & 0 \leq x < \frac{1}{2} \\ 4 - 4x, & \frac{1}{2} \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- The majoring function can be chosen as

$$g(x) = \begin{cases} 2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- $c = 2$ , and  $h(x) = \frac{g(x)}{2}$
- The acceptance condition becomes
  - $U \leq 2Y$ , if  $Y \leq \frac{1}{2}$
  - $U < 2 - 2Y$ , if  $Y \geq \frac{1}{2}$

# Example 1: Acceptance-Rejection



- $Y_i$ 's that fall near the middle part,  $f(x)$  is high
  - Most likely to be accepted
- Acceptance condition,  $Ut(Y) \leq f(Y)$  can visualize a point  $(Y, Ut(Y))$ 
  - If the point fall under the curve  $f(x)$ ,  $Ut(Y) \leq f(Y)$ 
    - $Y$  is accepted

# Example 2: Acceptance-Rejection

- Generate random variates of beta (4,3) distribution with PDF

$$f(x) = 60x^3(1 - x^2), \quad 0 \leq x \leq 1$$

- The majoring function is found by differentiating  $f(x)$

$$\begin{aligned} \frac{df(x)}{dx} &= 60x^2(5x^2 - 8x + 3) \\ &= 60x^2(5x - 3)(x - 1) \end{aligned}$$

- Now, setting  $\frac{df(x)}{dx} = 0$ , gives  $x = \frac{3}{5} = 0.6$
- The maximum value occurs at  $x = 0.6$ , which is 2.0736
- Majoring function,  $t(x) = 2.0736$ ,  $0 \leq x \leq 1$
- $c = \int_0^1 t(x)dx = 2.0736$
- $r(x) = 1, 0 \leq x \leq 1$

# Example 2: Acceptance-Rejection

- Algorithm
  - Step 1: Generate  $Y \sim U(0, 1)$
  - Step 2: Calculate  $z = \frac{60Y^3(1-Y)^2}{2.0736}$
  - Step 3: Generate  $u \sim U(0, 1)$ , independent of  $Y$
  - Step 4: If  $u \leq z$ ,  $X = Y$   
                    else go back to Step 1
  - Step 5: Return  $X$

# Composition Method

- The CDF of PDF of  $X$  can be expressed as
  - Convex combination of  $n$  CDF or PDF
- $F(x) = \sum_{i=1}^n p_i F_i(x)$        $f(x) = \sum_{i=1}^n p_i f_i(x)$ 
  - Where  $\sum_{i=1}^n p_i = 1$
- Algorithm
  - Step 1: Generate  $u_1 \sim U(0,1)$
  - Step 2: Use  $u_1$  to find the value of  $i$ , and the distribution  $F_i(x)$
  - Step 3: Generate  $u_2 \sim U(0,1)$
  - Step 4: Use  $u_2$  to generate  $X$  from  $F_i(x)$